

Decoupled 3-APK Native Specification & Android System Architecture

Native Android Service Decoupling, ADB Loopback & Sensory Streaming

Derived from: `What and Why.txt`, `Where and When.txt`, and `4-ARCHITECTURE-cont.-(6-files)`.
Status: Canonical System Architecture Spec | Clean Markdown Layer
Target Repository: `data_vault/02_wiki_md/`

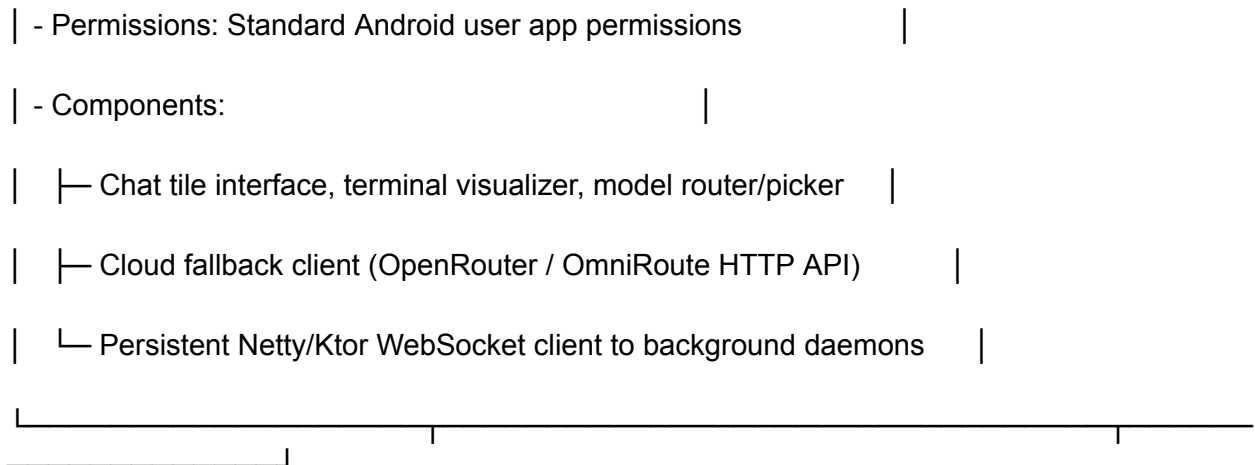
1. The Core Engineering Challenge & Solution

The Failure Mode of Monolithic Android Apps:
Running an LLM inference engine, speech pipeline, and reactive UI inside a single Android application process guarantees failure:

- 1. **Garbage Collection Freezes:** Model tokenization and inference introduce Java/Kotlin GC pauses that freeze UI rendering.
- 2. **Low Memory Killer (LMK) Panics:** Android aggressively terminates background processes that exceed memory thresholds (`FOREGROUND_APP_ADJ` boundaries).
- 3. **Sandbox Restrictions:** Standard Android applications cannot spawn persistent daemons, access hardware sockets directly, or maintain headless execution.

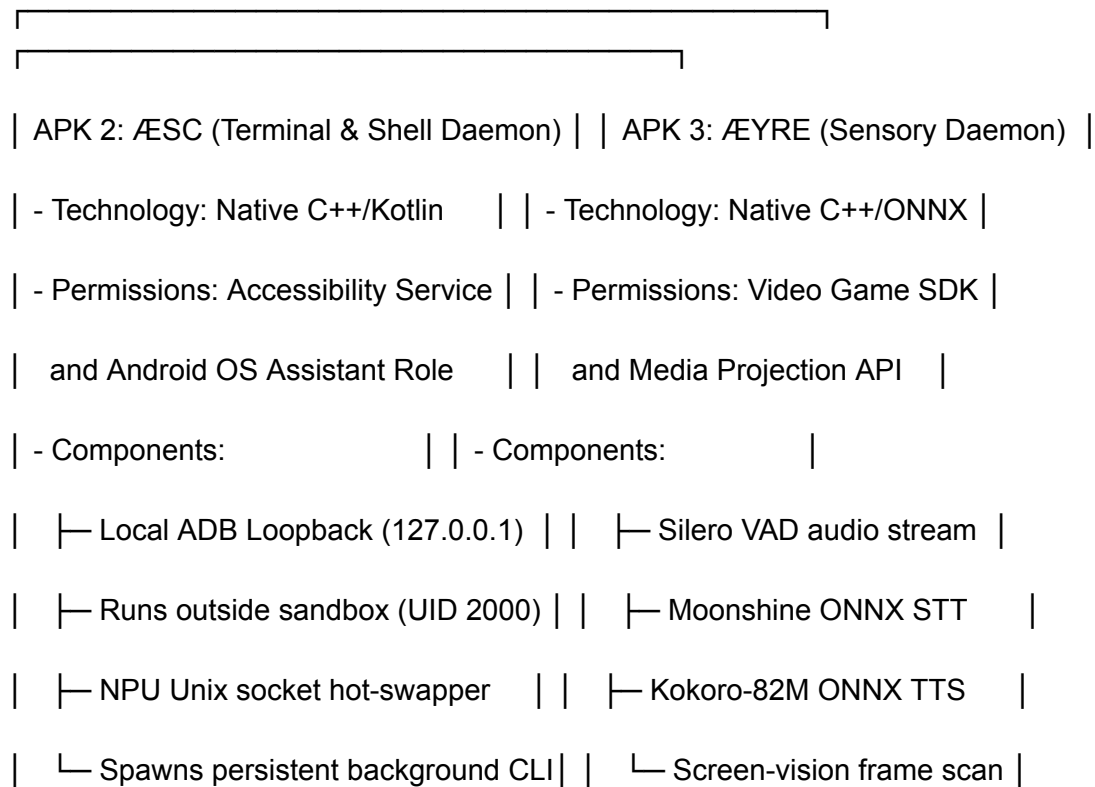
The Decoupled 3-APK Suite Architecture:
The system bypasses Termux and standard application sandboxes by partitioning on-device operations into three native APKs communicating over local abstract UNIX sockets (`\0`) and WebSockets (`127.0.0.1:8080`, `127.0.0.1:8765`):

	APK 1: HORIZONS UI (Master Presentation Shell & Concierge)	
	- Technology: Chromium WebView (HTML5 / React user interface)	



UNIX Abstract Socket / WebSocket UNIX Abstract Socket / WebSocket

/dev/socket/aesc_shell.sock /dev/socket/aeysre_media.sock



2. The Local ADB Loopback Mechanism (Zero-Root Privilege Escalation)

- **UID Separation:** Standard apps run with isolated UIDs (e.g. `u0_a245`). The Android `adb` daemon runs with UID 2000 (`shell`).
 - **Wireless Debugging Bridge:** By enabling Wireless Debugging, Android exposes `adb` over a local TCP loopback interface on `127.0.0.1`.
 - **Execution Flow:**
 1. **Pairing:** `Æsc` generates an RSA key pair (`~/.android/adbkey`) and connects once via `adb pair 127.0.0.1:<pairing_port>`.
 2. **Connection:** On every system boot (`boot.sh`), `Æsc` automatically issues `adb connect 127.0.0.1:<port>`.
 3. **Privilege Escalation:** Commands launched through this local loopback execute as UID 2000 (`shell`), enabling persistent background processes, socket creation, and file manipulation across app boundaries without rooting the device.
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3. The Concierge Processing Sequence

[User Speaks / Screen Event]

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[Æyre Media Daemon] —► Silero VAD detects voice → Moonshine STT transcribes speech

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Screen Vision captures active frame context



[Horizons UI Concierge] —► Passes stream to 0.8B Qwen model to strip run-on typos

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Compiles clean, structured Markdown meta-prompt



[User One-Tap Verification] (Optional prompt review on screen tile)

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[OmniRoute Dispatcher] (Port 20128) —► Local NPU (9B) OR Cloud Frontier (Claude/Vertex)

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[Model Completion] —► Kokoro TTS synthesizes voice response to earbud

Active execution steps logged to Reasoning Bank